

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- Magnetic spark plug socket, Part Number 3824876

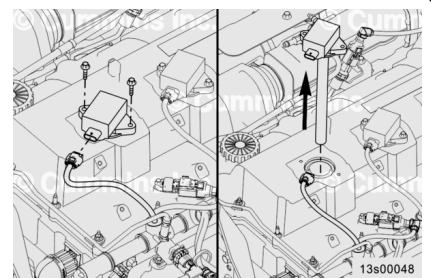
Additional Service Items

- Wire spark plug gap gauge.

Remove

⚠ WARNING ⚠

To reduce the possibility of personal injury, do not touch any ignition wires or components while the engine is operating, unless using suitably insulated tools.



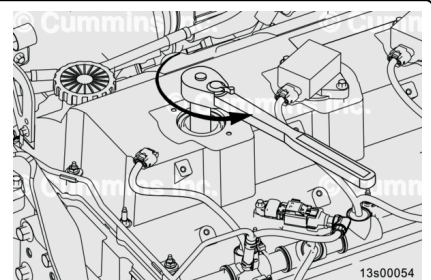
Remove the ignition coil. Refer to Procedure 013-012 in Section 13. (/qs3/pubsys2/xml/en/procedures/1016/1016-013-012.html)

Blow out any dirt and debris from the cylinder head and spark plug adapter tube.

Use a non-magnetic tapered 5/8 inch deep well socket, such as a standard deep well 5/8 inch spark plug socket without a rubber insert, to remove the spark plug.

Loosen the spark plugs. Use a 5/8 inch magnetic spark plug socket, Part Number 3824876.

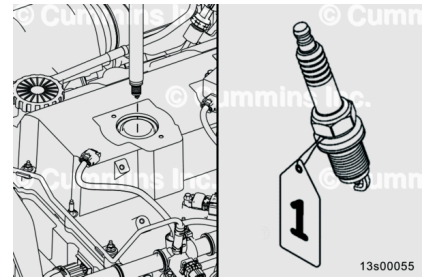
Do **not** use a spark plug socket with a rubber insert.



Use a magnetic spark plug socket and continue to back out the spark plug until the threads are no longer engaged.

Lift the spark plug out with the socket, being careful **not** to drop the plug.

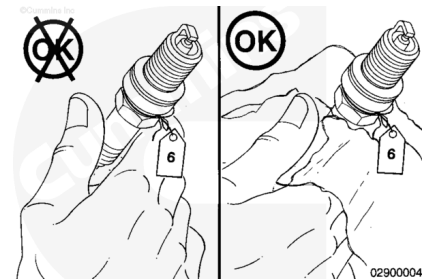
If the spark plugs are to be reused, carefully mark or tag the spark plug according to the cylinder from which it is removed. Do **not** mark the ceramic or damage the electrode.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

⚠ CAUTION ⚠

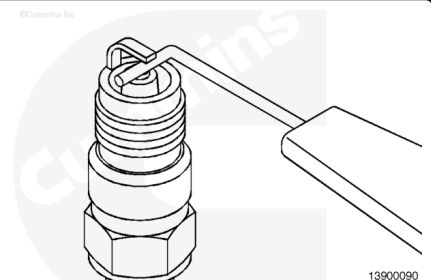
Spark plug life is largely dependent on cleanliness of the plug's porcelain. Do not touch the porcelain area on the spark plug.

Note : A convenient option is an alcohol prep pad that does **not** contain any moisturizers or other cleaners. Compliant alcohol prep pads are available at many department stores, drug stores, and industrial supply centers.

⚠ CAUTION ⚠

Damage to the electrodes due to improper use of a feeler gauge will shorten the life of the spark plug.

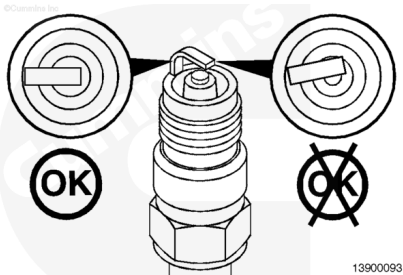
Clean the tool gauges prior to usage.



Use a wire spark plug gap gauge to check the spark plug gap.
Replace the spark plug if out of specification.

Gap Specification

	mm		in
New spark plug	0.35	MIN	0.014
	0.40	MAX	0.016
Reused spark plug	0.35	MIN	0.014
	0.60	MAX	0.024



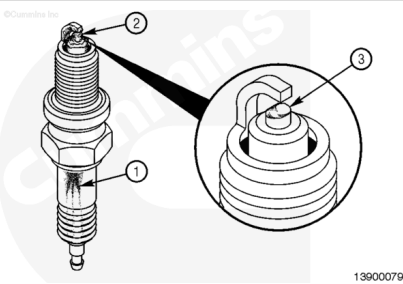
Final settings: Do **not** adjust gap.

Check the spark plug. The ground electrode **must** be parallel to the center electrode and **must** be aligned along the centerline of the plug. Replace spark plug if out of specification.

Refer to Service Bulletin, Spark Plug Reuse Guidelines, Bulletin 4383578 (/qs3/pubsys2/xml/en/bulletin/4383578.html), to determine if the spark plug meets reuse guidelines.

Inspect the spark plugs for the following issues:

1. Insulator flashover
2. Electrode deposits or fouling
3. Electrode misalignment
4. Missing electrode (**not** shown)
5. Cracked insulator (**not** shown).



A worn electrode does **not** mean that the spark plug **must** be replaced.

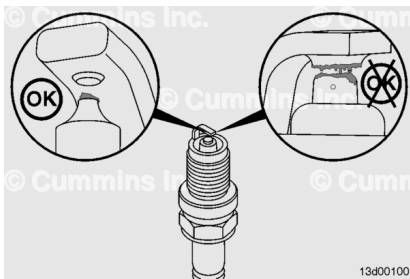
If a spark plug is found with melted or damaged electrodes, it can indicate piston damage. Inspect the piston for damage. See equipment manufacturer service information.

Corona staining does **not** affect spark plug or ignition system performance.

Spark plugs shall **not** be replaced due to corona staining (3).

Corona staining is the light brown discoloration found on the ceramic portion, just above the metal shell.

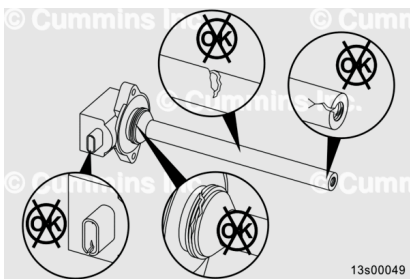
If a spark plug exhibits electrode beading, check the electrode gap before discarding the plug. Replace the plug **only** if the gap is less than 0.330 mm [0.013 in].



Refer to Service Bulletin, Ignition Coil Malfunctions, Bulletin 4383630 (/qs3/pubsys2/xml/en/bulletin/4383630.html), to determine if the coil extension is reusable.

Inspect the coil extension for physical damage. Refer to Procedure 013-050 in Section 13.

(/qs3/pubsys2/xml/en/procedures/1016/1016-013-050-tr.html)



Install

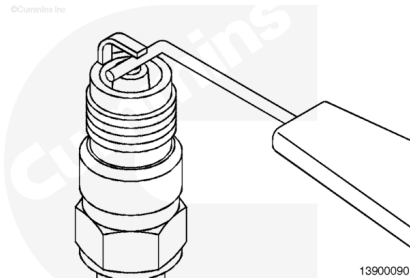
New Spark Plugs

⚠ CAUTION ⚠

Damage to the electrodes due to improper use of a feeler gauge will shorten the life of the spark plug.

Clean the tool gauges prior to usage.

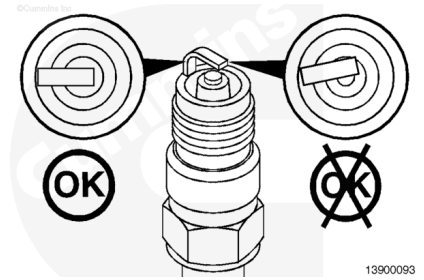
Use a wire spark plug gap gauge to check spark plug gap.
Replace spark plug if out of specification.



Gap Specification			
	mm		in
New spark plug	0.35	MIN	0.014
	0.40	MAX	0.016
Reused spark plug	0.35	MIN	0.014
	0.60	MAX	0.024

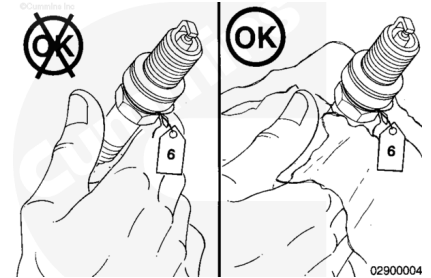
Final settings: Do **not** adjust gap.

Check the spark plug. The ground electrode **must** be parallel to the center electrode and **must** be aligned along the centerline of the plug. Replace spark plug if out of specification.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

⚠ CAUTION ⚠

Spark plug life is largely dependent on cleanliness of the plug's porcelain. Do not touch the porcelain area on the spark plug.

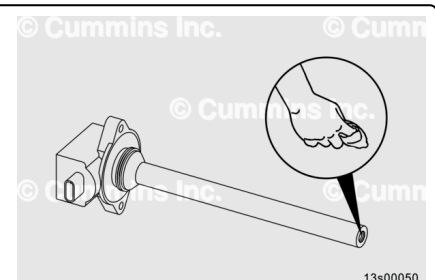
Clean the magnetic spark plug socket and spark plug porcelain. Use rubbing alcohol and a lint-free cloth.

Note : A convenient option is an alcohol prep pad that does not contain any moisturizers or other cleaners. Compliant alcohol prep pads are available at many department stores, drug stores, and industrial supply centers.

Before installing the spark plugs, check inspection criteria above for the spark plug and coil extension.

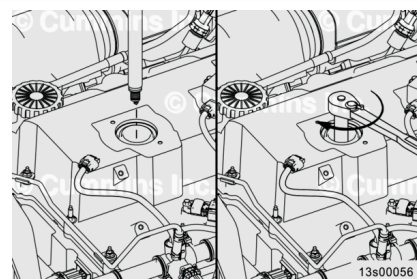
Use a clean, lint-free cloth to remove any debris or residue from inside the end of the coil extension.

Use a cotton swab and isopropyl alcohol to remove any excess dielectric grease from the spark plug end of the coil extension spring.



⚠ CAUTION ⚠

Mechanical over-tightening can damage the spark plug and cylinder head.



⚠ CAUTION ⚠

Proper spark plug torque is required for heat transfer from the spark plug to the cylinder head. Improper torque will lead to spark plug malfunction, and possible engine damage.

Inspect the cylinder head and spark plug adapter tubes for debris. Clean as required.

Install the spark plug.

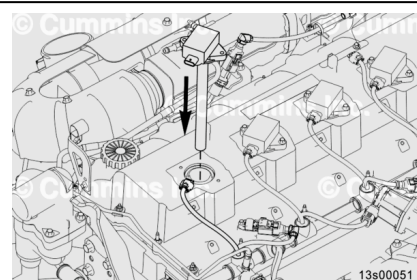
Use a magnetic spark plug socket, Part Number 3824876, and thread the plug in by hand to make sure that the threads are smooth and the plug is **not** cross-threaded.

Tighten the spark plug.

Torque Value: 36 N·m [27 ft-lb]

Note : For applications with limited overhead clearance, use a collapsible extension.

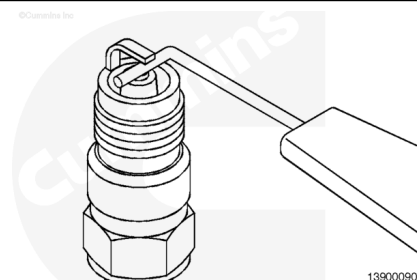
Install the ignition coil and coil extension. Refer to Procedure 013-012 in Section 13.
(/qs3/pubsys2/xml/en/procedures/1016/1016-013-012.html)



Reinstalling Spark Plugs

⚠ CAUTION ⚠

Damage to the electrodes due to improper use of a feeler gauge will shorten the life of the spark plug.

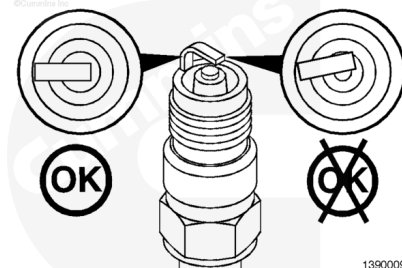


Gap Specification			
	mm		in
New spark plug	0.35	MIN	0.014
	0.40	MAX	0.016
Reused spark plug	0.35	MIN	0.014

	0.60	MAX	0.024
--	------	-----	-------

Final settings: Do **not** adjust gap.

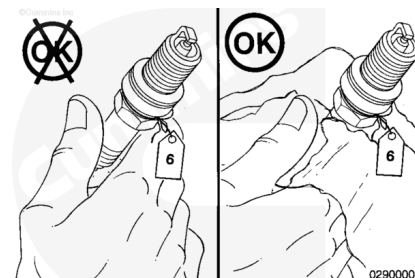
Check the spark plug. The ground electrode **must** be parallel to the center electrode and **must** be aligned along the centerline of the plug. Replace spark plug if out of specification.



13900093

⚠ CAUTION ⚠

Spark plug life is largely dependent on cleanliness of the plug's porcelain. Do not touch the porcelain area on the spark plug.



02900004

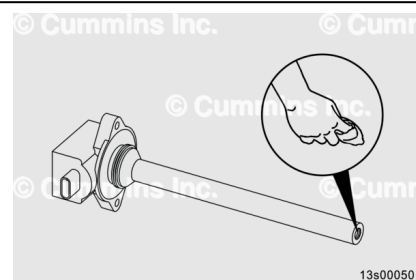
Clean the magnetic spark plug socket and spark plug porcelain. Use rubbing alcohol and a lint-free cloth.

Note : A convenient option is an alcohol prep pad that does **not** contain any moisturizers or other cleaners. Compliant alcohol prep pads are available at many department stores, drug stores, and industrial supply centers.

Before installing the spark plugs, check inspection criteria above for the spark plug and coil extension.

Use a clean, lint-free cloth to remove any debris or residue from inside the end of the coil extension.

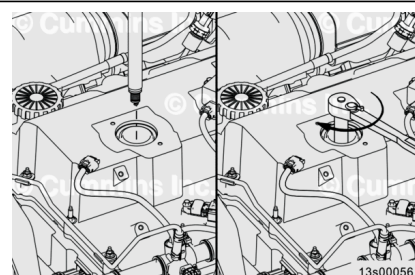
Use a cotton swab and isopropyl alcohol to remove any excess dielectric grease from the spark plug end of the coil extension.



13s00050

⚠ CAUTION ⚠

Mechanical over-tightening can damage the spark plug and cylinder head.



13s00056

⚠ CAUTION ⚠

Proper spark plug torque is required for heat transfer from the spark plug to the cylinder head. Improper torque will lead to spark plug malfunction and possible engine damage.

Inspect the cylinder head and spark plug adapter tubes for debris. Clean as required.

Install the spark plug and hand tighten.

Use a magnetic spark plug socket, Part Number 3824876, and thread the plug in by hand to make sure that the threads are smooth and the plug is **not** cross-threaded.

Tighten the spark plug.

Torque Value: 36 n•m [27 ft-lb]

Note : For applications with limited overhead clearance, use a collapsible extension.

Install the ignition coil and coil extension. Refer to Procedure 013-012 in Section 13.

(/qs3/pubsys2/xml/en/procedures/1016/1016-013-012.html)

